GIS TERM PROJECT · FINAL PRESENTATION

Geospatial Reasoning using LLMs and Generative AI

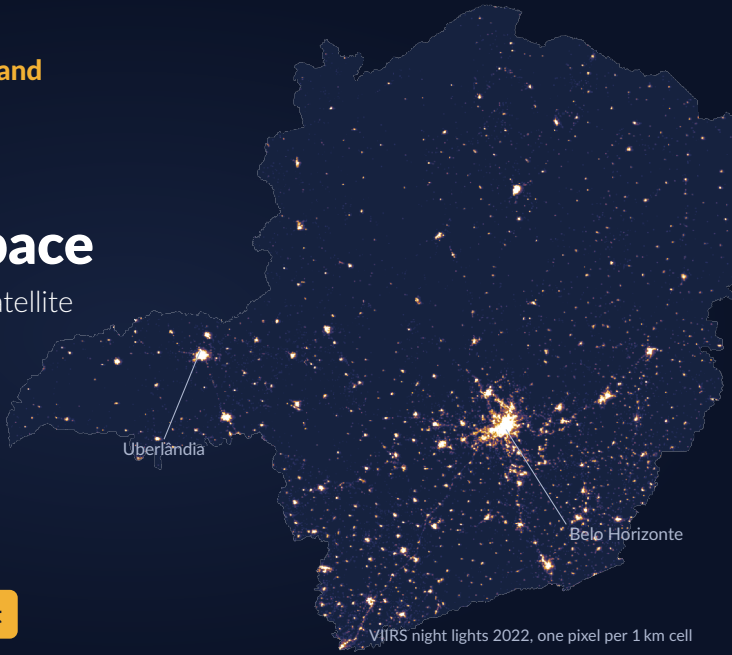
Reading the economy from space

Estimating municipal GDP from satellite embeddings, text and night lights

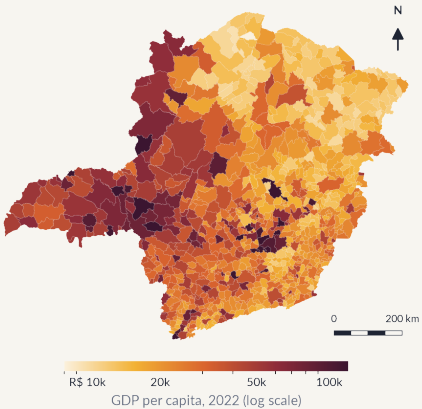
Group 2

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github.com/chadsaras/GIS_Project



Local GDP is patchy; satellites see every place



49×

gap in GDP per capita inside one state:

Catas Altas (R$ 391k) vs São João das Missões (R$ 8k)

- Sub-national GDP is published **late**, **estimated** from partial data, or **missing** where statistics are weak.
- Satellites measure **every place, every year, with the same sensor**.

GOAL

Estimate GDP per capita for all **853 municipalities** of Minas Gerais from **free satellite data**, and test it **honestly**.

Two recent ideas, never tested together

UrbanCLIP Yan et al., WWW 2024

A language model writes a **caption for every satellite tile**; image and text are aligned.

– GDP gain only +0.5% R^2; random split lets neighbours leak into the test

Satellite embeddings Yue, Zhao & Hu, 2026

Google's **64-number embeddings**, **steered by night lights** towards economic meaning.

– countries only; no text, no POI or roads

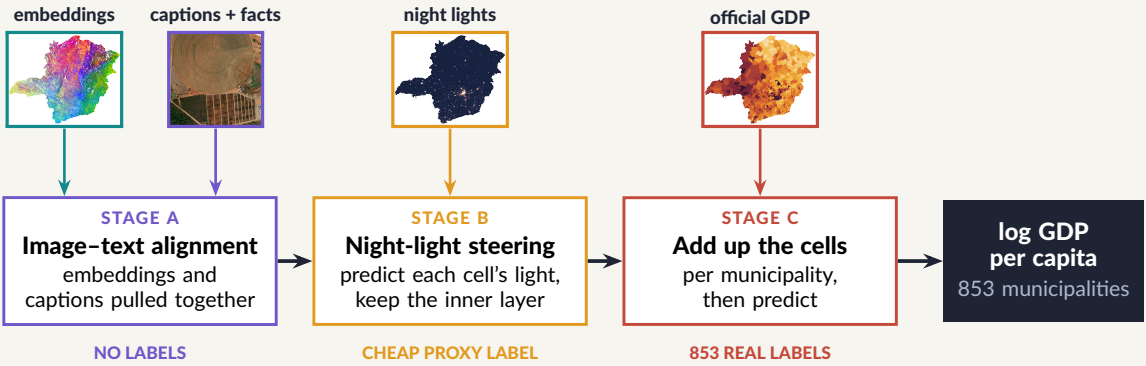
THIS PROJECT

Both, in one pipeline

● embeddings ● text
● night lights ● POI, roads

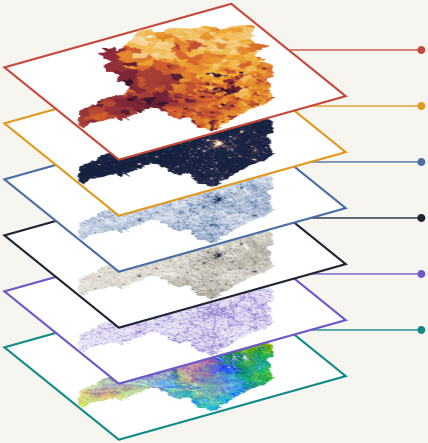
853 municipalities · spatial test
ablation of every ingredient

Three learning stages, from no labels to official GDP



Captions are needed only in training: at prediction time each cell uses its embedding alone.

Six free layers on one 1 km grid



GDP per capita (target)
IBGE 2022, official

Night lights
VIIRS 2022

Roads and POIs
OpenStreetMap

Buildings
Microsoft + OSM footprints

Captioned tiles
10,000 Esri tiles, Qwen2.5-VL

Satellite embeddings
Google GSED, 64 numbers, 10 m to 1 km

853
municipalities

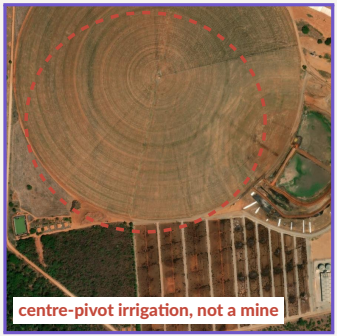
575k
1 km cells

10,000
captioned tiles

10M
building footprints

All layers free and aligned on the same grid (EPSG:31983). The target is official statistics, not night lights.

An AI model describes each tile; map data vetoes false claims



Esri World Imagery, about 1 m per pixel

QWEN2.5-VL-7B CAPTION, AFTER THE CHECKS

- ✗ ~~The satellite image shows a large circular area with concentric rings, likely a mine pit, surrounded by a grid of smaller rectangular plots of farmland.~~ **no industry in the data**
- ✓ Roads crisscross the area, connecting the plots and the perimeter.
- ✓ Vegetation is sparse, mostly along the edges and in some patches within the farmland.
- ✓ A small body of water, possibly a pond, is visible near the farmland.
- ✓ No buildings or industrial structures are evident, suggesting minimal built-up area.
- ✗ ~~The overall scene appears to be an agricultural landscape with a focus on farming and possibly mining operations.~~ **filler**

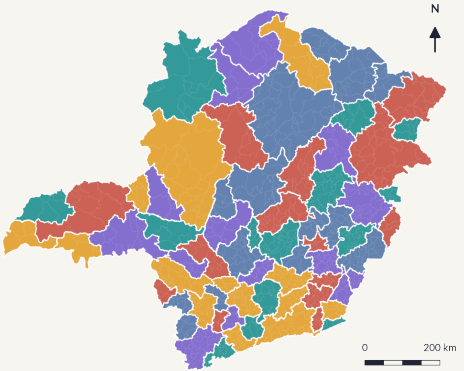
SENTENCES REMOVED, BY RULE

Filler, no content	4,183
Industry / mine	1,130
Building	1,108
Water	1,046
Road	1,039
Farmland	732

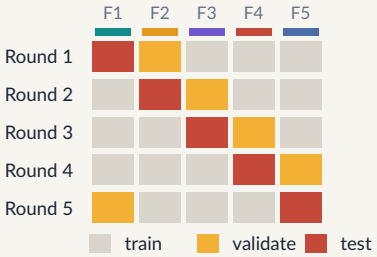
84% kept of 60,023 sentences

Checked against ESA land cover, OSM roads and building footprints; image–text matching removes 246 more captions.

Every municipality is tested in a region never seen



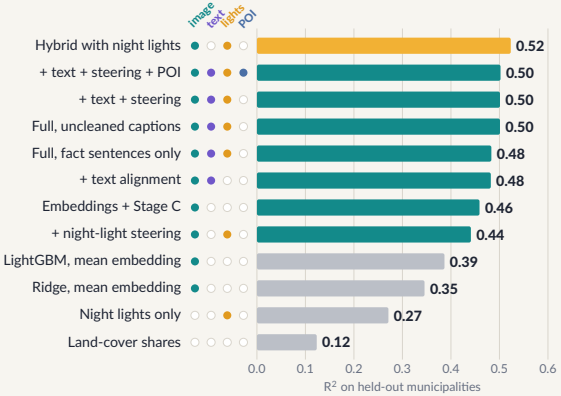
70 official regions (white outlines) grouped into 5 folds



- Whole regions held out: **no look-alike neighbour** in training. All stages retrained each round.
- UrbanCLIP split tiles **at random**, so neighbours sat on both sides and scores were inflated.

Models are compared on the same folds with a block bootstrap over regions (95% intervals).

Text gives the biggest gain; the hybrid explains half of GDP



Dots: ingredients each model uses. Hybrid = best model blended with night lights.

0.52 R^2, hybrid, unseen regions

RQ2 TEXT **largest gain, borderline**
ΔRMSE -0.025, 95% CI [-0.050, +0.004]

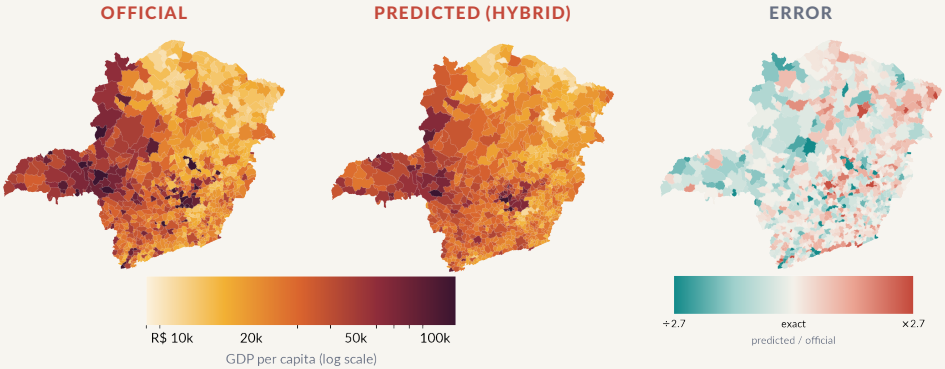
RQ1 STEERING **no clear effect**
ΔRMSE +0.007, 95% CI [-0.028, +0.041]

RQ3 POI, ROADS **nothing added**
ΔRMSE -0.001, 95% CI [-0.017, +0.016]

RQ4 HYBRID **best, not significant**
ΔRMSE -0.009, 95% CI [-0.035, +0.015]

RMSE differences on log GDP per capita with 95% block-bootstrap intervals; negative = first model better.

Predictions follow the map, but are pulled towards the middle



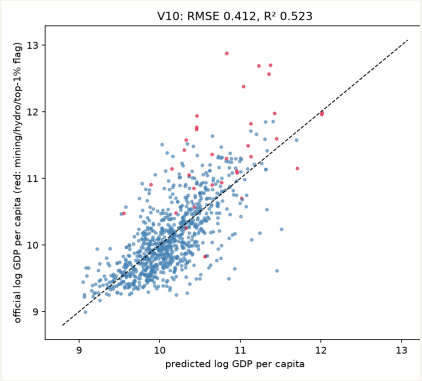
75%

of municipalities predicted within a factor 1.5 of official

Under-predicted: rich agribusiness and mining towns (west, centre).

Over-predicted: small public-sector and commuter towns.

The biggest misses are mines and steel mills satellites cannot see



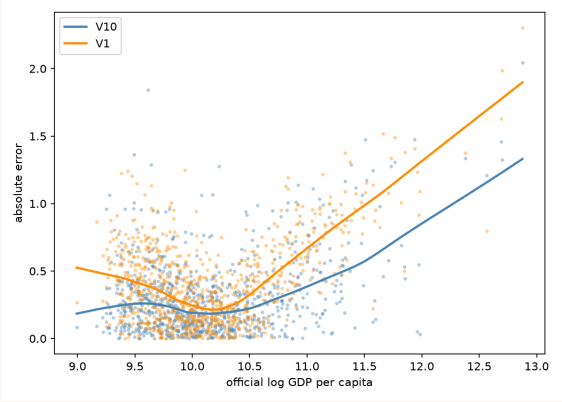
Hybrid, 853 test predictions; red = flagged mining or hydro.

TEN LARGEST GAPS

Municipality	gap	likely reason
Catas Altas	−2.04	iron-ore mine
Santa Cruz de Minas	+1.84	tiny commuter town
Ressaquinha	−1.48	large plant
Conceição do Mato Dentro	−1.48	iron-ore mine
S. Gonçalo do Rio Abaixo	−1.46	iron-ore mine
Mário Campos	+1.37	BH commuter town
Itatiaiuçu	−1.34	iron-ore mine
Extrema	−1.32	logistics hub
Ouro Branco	−1.31	steel mill
Chiador	+1.29	public-sector town

Gap = predicted − official (log). A few cells make most of the output. Reasons are candidates.

Errors grow with income, but half as fast as with night lights



Absolute error against official income, LOWESS. Blue: hybrid; orange: night lights only.

0.08 vs **0.19**

extra error per log point of income, hybrid vs night lights

0.16 vs **0.33**

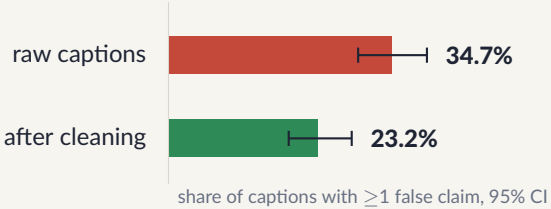
Moran's I of errors: still clustered, half as much

$\rho = 0.04$

ensemble spread does not predict error: seeds are not a confidence interval

Error regression with region-clustered SEs: errors rise with income and built-up share, fall with services share; OSM completeness does not matter.

Data checks cut false captions from a third to a quarter



CLEANING

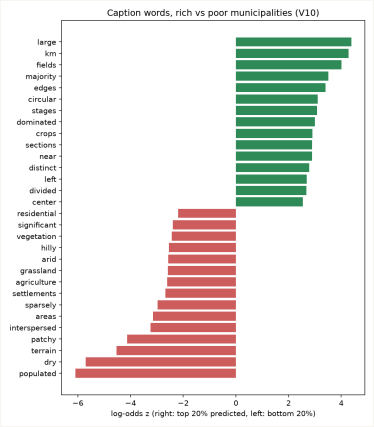
removes **41%** of false sentences; **58%** of what it removes is false or vague

TYPICAL FALSE CLAIMS

- “no buildings” next to a few farmhouses
- “farmland” on natural scrub or forest
- “mining activity” on centre-pivot fields
- ponds and roads that are not there

300 random captions, 1,801 sentences labelled correct / false / vague, blind to the cleaning. Labelled by an LLM (Claude) looking at each tile, not by people; a human check is still open. Noise did not hurt GDP: uncleaned captions (V8) did as well as cleaned (V7).

The text space speaks the state's economic geography



RICHEST 20% PREDICTED

large, *fields*, *circular*, *crops*, *divided*: mechanised farming and centre pivots of the Triângulo

POOREST 20% PREDICTED

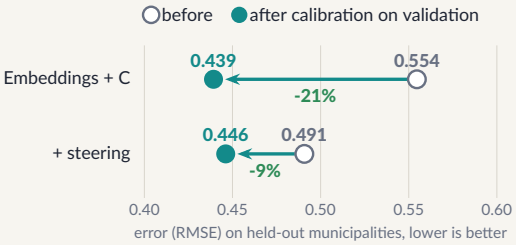
sparsely populated, *dry*, *terrain*, *patchy*, *arid*: the semi-arid north and Jequitinhonha

NEAREST TRAINING CAPTIONS

Belo Horizonte (10.82 vs 10.91): dense grids of red-tiled roofs.
Conceição do Mato Dentro (11.94 vs 10.46): forest and pasture, so the mine stays invisible.

Log-odds with informative Dirichlet prior (Monroe et al., 2008); official vs predicted log GDP per capita.

Calibration mattered more than night-light steering



1. **Predictions were over-spread.** A straight-line correction fitted on validation municipalities cut error by **−21%**.
2. **Honest testing changes conclusions.** Steering looked helpful before the fix; after it, the gain vanished.
3. **Captions need checking.** The model invents mines, roads and farmland on plain tiles; simple data checks remove four in ten false sentences.

Answers to the five questions, and what we could not do

RESEARCH QUESTIONS

RQ1 night-light steering	NO CLEAR EFFECT
RQ2 text (captions)	LARGEST GAIN, BORDERLINE
RQ3 POIs and roads	NOTHING ADDED
RQ4 hybrid	BEST, NOT SIGNIFICANT
RQ5 where it fails	MINES, STEEL, COMMUTER TOWNS

Every answer out-of-sample on regions never seen; 95% block-bootstrap intervals.

LIMITS

- GSED is frozen: text can reorganise, not add, image information.
- Caption audit labelled by an LLM, one labeller, no agreement score yet.
- Stage C was revised after the first test results (choices made on validation).
- Growth over 2018–2021 not tested: Earth Engine quota ran out.
- One state only; transfer elsewhere untested.

TAKE-AWAY

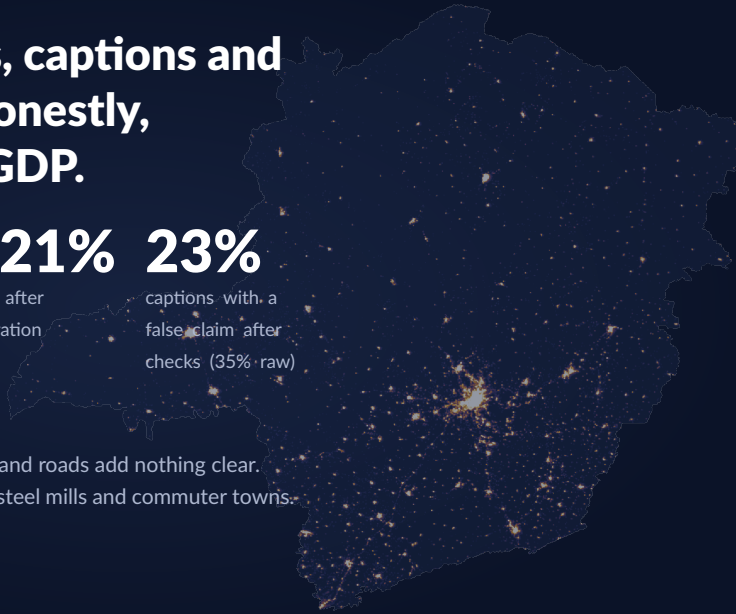
Satellite embeddings, captions and night lights, tested honestly, explain half of local GDP.

0.52
R^2, hybrid, unseen regions

0.27
R^2, night lights alone

−21%
error after calibration

23%
captions with a false claim after checks (35% raw)



Captions give the largest gain; steering, POIs and roads add nothing clear.
Satellites miss what they cannot see: mines, steel mills and commuter towns.

Questions?

github.com/chadsaras/GIS_Project